

**CMR COLLEGE OF ENGINEERING & TECHNOLOGY**
(UGC AUTONOMOUS)

B.Tech II Semester Supplementary Examinations February-2024

Course Name: ODE'S AND MULTIVARIABLE CALCULUS
(Common for all Branches)**Date: 15.02.2024 FN****Time: 3 hours****Max.Marks: 70**

(Note: Assume suitable data if necessary)

PART-A

Answer all TEN questions (Compulsory)

Each question carries TWO marks.

10x2=20M

1. Find the integrating factor of $x \frac{dy}{dx} + y = \log x$ 2 M
2. Solve $x^2 - yp + a = 0$ 2 M
3. Solve the differential equation $(D^2 + 5D + 6)y = 0$ 2 M
4. Define Cauchy-Euler equation. 2 M
5. Find $\int_0^2 \int_0^x e^{x+y} dy dx$ 2 M
6. Evaluate $\int_0^1 \int_0^2 \int_0^3 xyz \, dx dy dz$ 2 M
7. Find the gradient of scalar function $\phi(x, y, z) = 3x^2 y + y^3 z^2$ at $(1, -2, -1)$. 2 M
8. Prove that $\text{curl grad } \phi = 0$. 2 M
9. Evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = x^2 \vec{i} + y^2 \vec{j}$ and C is the curve $y = x^2$ in the xy-plane 2 M
from $(0,0)$ to $(1,1)$
10. State Gauss Divergence theorem 2 M

PART-B**Answer the following. Each question carries TEN Marks.****5x10=50M**

- 11.A). i) Solve the differential equation $(x^2 + 2y^2) dx - xy dy = 0$ 5M
ii) Solve the differential equation $x \frac{dy}{dx} + y = x^3 y^6$ 5M
- OR**
11. B). A body kept in air with temperature 25°C cools from 140°C to 80°C in 20 minutes. Find when the body cools down to 35°C . 10M
12. A). Solve $(D^2 + 4)y = \tan 2x$ by method of variation of parameter. 10M
- OR**
12. B). Solve $x^3 \left(\frac{d^3 y}{dx^3} \right) + 3x^2 \left(\frac{d^2 y}{dx^2} \right) + x \left(\frac{dy}{dx} \right) + 8y = 65 \cos(\log x)$ 10M

(P.T.O.)

13. A). Change of order of integration and hence evaluate the double integral $\int_0^1 \int_{x^2}^{2-x} xy dy dx$ 10M

OR

13. B). Find the volume of the tetrahedron bounded by the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ and the coordinate planes by triple integral. 10M

14. A). i) Find the directional derivative of $xyz^2 + xz$ at $(1,1,1)$ in the direction of normal to the surface $3xy^2 + y = z$ at $(0,1,1)$. 5M

ii) Show that $(x^2 - yz)\bar{i} + (y^2 - zx)\bar{j} + (z^2 - xy)\bar{k}$ is irrotational and find its scalar potential. 5M

OR

14. B). Prove that $\text{curl}(\bar{a} \times \bar{b}) = \bar{a} \text{div} \bar{b} - \bar{b} \text{div} \bar{a} + (\bar{b} \cdot \nabla)\bar{a} - (\bar{a} \cdot \nabla)\bar{b}$. 10M

15. A). Find the work done by $\bar{F} = (2x - y - z)\bar{i} + (x + y - z)\bar{j} + (3x - 2y - 5z)\bar{k}$ along a curve C in the xy-plane given by $x^2 + y^2 = 4, z = 0$. 10M

OR

15. B). Verify Green's theorem for $\bar{F} = (x^2 + y^2)\bar{i} - 2yx\bar{j}$ taken around the rectangle bounded by the lines $x = a, x = -a, y = 0, y = b$. 10M
